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Automatic feature extraction and selection for classification of cyclical time series data

Automatische Merkmalsextraktion und -selektion von Merkmalen zur Klassifikation zyklischer Signalverläufe

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Abstract: The classification of cyclically recorded time series plays an important role in measurement technologies. Example use cases range from gas sensors combined with temperature cycled operation to condition monitoring using vibration analysis. Before machine learning can be applied to high dimensional cyclical time series data dimensionality reduction has to be performed to avoid the classifier suffering from overfitting and the “curse of dimensionality”. This paper introduces a set of four complementary feature extraction methods and three feature selection algorithms that can be applied in a fully automated manner to reduce the number of dimensions. The feature extraction algorithms are capable of extracting characteristic features from cyclical time series catching information contained in local details and overall cycle shape as well as in frequency or time-frequency domain. The methods for feature selection are capable of selecting the most suitable features for linear and nonlinear classification. The methods were chosen to be applicable to a wide range of applications which is verified by testing the set of methods on four different use cases.

Keywords: Time series classification, dimensionality reduction, machine learning.

Zusammenfassung: Die Klassifikation zyklischer Signalverläufe mittels maschinellen Lernens spielt eine wichtige Rolle in der Messtechnik. Beispielanwendungen reichen von Gas-Sensoren, die temperaturzyklisch betrieben werden, bis hin zur Zustandsüberwachung durch Vibrationsanalyse. Bevor maschinelles Lernen auf die hochdimensionalen, zyklischen Signalverläufe angewandt werden kann, muss deren Dimensionalität verringert werden, um zu verhindern, dass der Klassifikator unter Overfitting und dem „curse of dimensionality“ leidet. In dieser Veröffentlichung werden vier sich gegenseitig ergänzende Methoden zur Merkmalsextraktion und drei Algorithmen zur Merkmalselektion vorgeschlagen, die automatisiert genutzt werden können, um die Dimensionalität zu verringern. Die Algorithmen zur Merkmalsextraktion extrahieren charakteristische Merkmale aus dem Signalverlauf. Die in den Merkmalen enthaltene Information beinhaltet dabei nicht nur lokale Details und die allgemeine Kurvenform sondern auch Merkmale aus dem Frequenz- und Zeit-Frequenz-Bereich. Die Methoden zur Merkmalselektion sind in der Lage die besten Merkmale für lineare und radiale Klassifikation auszuwählen. Die Methoden wurden so ausgewählt, dass sie für ein möglichst breites Anwendungsspektrum geeignet sind, was durch die erfolgreiche Anwendung auf vier verschiedene Beispieldatensätze gezeigt wird.

Schlüsselwörter: Klassifikation von Signalverläufen, Dimensionsreduktion, maschinelles Lernen.

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1 Introduction

Time series play an important role in measurement technologies. Thereby cyclical time series, which are for example recorded during a fixed working cycle of an industrial machine, are of special interest. Classification of such signals can be used for condition monitoring of the machine. If machine faults have different effects during different parts of the machine working cycle the fault mechanism

produces a certain “fingerprint” in the cyclical sensor data that can be used to determine the fault progress allowing predictive maintenance [1, 2]. Using machine learning trained on a database containing cycles coming from machines with and without faults these fingerprints can be identified. The problem addressed in this paper is that common machine learning algorithms are not suitable for time series classification due to the high dimensionality of raw data, if each measurement point in a cycle is treated as a new dimension. One issue is the computational complexity of the learning algorithm making learning in high-dimensional space inefficient. More important, however, are effects like overfitting and the “curse of dimensionality” [3] which reduce classifier performance in high-dimensional space.

2 Dimensionality reduction

To apply machine learning to high-dimensional data, suitable dimensionality reduction has to be performed by extracting characteristic features from the time series (feature extraction) and selecting the most relevant features (features selection). Feature extraction is an unsupervised (i.e. is done without knowledge of the cycle’s group affiliation) process that only depends on the basic shape of the cycle which should be represented as accurate as possible by the features. However, feature selection selects the most relevant features with respect to a given classification target. Therefore, feature selection is a supervised process which depends on knowing the correct group affiliation of each cycle.

To reduce the effort spent on dimensionality reduction, this paper suggests a complementary set of automated methods for feature extraction and selection. Since there is no universal method for dimensionality reduction of time series a set of algorithms is better suited to cover a wide range of possible applications as a single algorithm. The methods suggested for feature selection were chosen for working best on different applications of current research, i.e. by offering complementary performance. Additionally, the individual algorithms also have to be as effective and efficient as possible. Efficiency in the context of time series means being scalable, both with respect to a high number of data points per cycle, to account for long cycles and high sampling rates, and a high number of cycles, to account for large databases. Effectiveness of feature extraction is the ability of the algorithm to represent the original cycle as accurately as possible using as few relevant features as possible. In the process the algorithm must not create new clusters within the database because

new clusters increase model complexity needed to learn classification rules for fault types. Creating new clusters is a typical risk when using feature extraction methods that use features for bookkeeping. With respect to the consecutive feature selection the feature extraction algorithm should concentrate the relevant information in few features to allow feature selection to easily identify and select the most relevant features.

3 Feature extraction

Each of the four feature extraction algorithms described in this chapter is suitable for representing different types of information typically contained in time series data. In this sense the algorithms complement each other to process data from a broad spectrum of real world applications.

The four methods suggested for feature extraction after comparison of a total of 15 methods [4] are:

3.1 Adaptive linear approximation (ALA)

Information contained in local details like transients and edges can be represented by piecewise approximation of the cycle with linear functions. Piecewise approximation splits the cycle into several variable length segments which are represented by the linear fit parameters mean value and slope. This is particularly useful if relevant information is only contained in certain cycle segments because relevant information is concentrated in few features allowing feature selection to safely discard irrelevant information. The linear fit function is chosen because it is computationally cheap and provides first order approximations. Since irrelevant signal components like noise require higher order approximations they are efficiently suppressed. In addition, multiple automated algorithms are available for linear segmentation. The only algorithm which guarantees minimal regression error over all cycles for a given number of segments is described in [5]. This algorithm also automatically suggests a reasonable number of segments to use and, thus, automatically tunes the tradeoff between low approximation error and low number of features. Figure 1(l) shows the resulting approximation of the a gas sensor response to TCO.

3.2 Principal Component Analysis (PCA)

The best way to catch the general trend of a cycle in the time domain is to use the projection onto the first principal

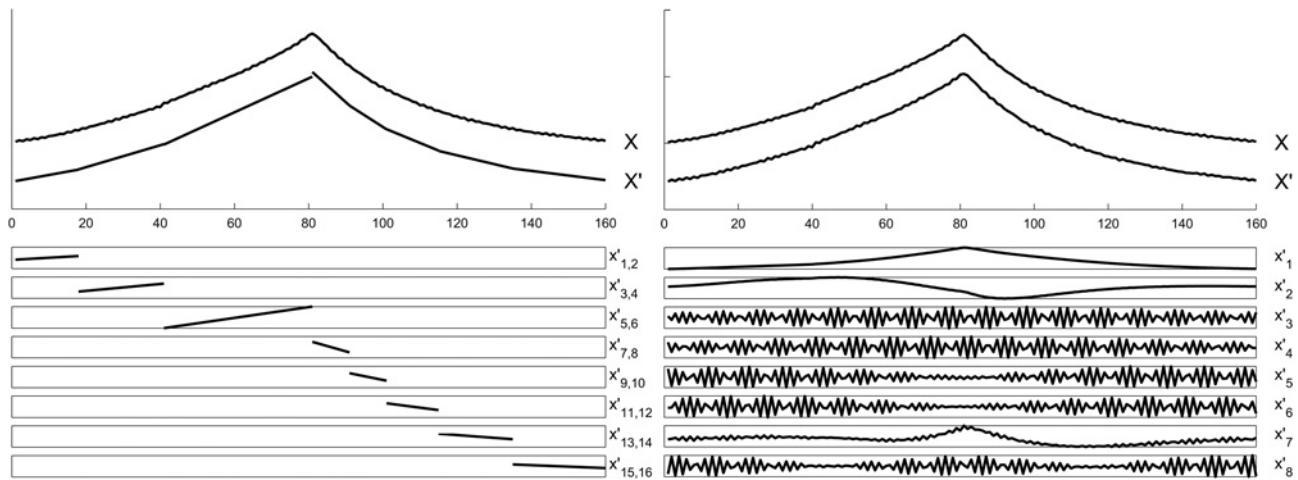


Figure 1: (l) Approximation X' using ALA with features x'_1 – x'_{16} , compared to the original gas sensor temperature cycle response X (shifted for better clarity). The eight segments are represented by mean values (uneven indices) and slopes (even indices). (r) Approximation using PCA. Extracted features x'_1 – x'_8 are projections onto the first principal components that are shown below.

components (PCs) found by PCA. This method is chosen because the first principal components are the best linear transformation of the dataset for a given number of features (projections on PCs) in terms of approximation error [6]. An alternative view to PCA is that the first PCs are formed to explain maximum variance and therefore capture fundamental cycle trends while later PCs capture details and noise. This enables the PCA to determine the number of PCs required to represent all non-random variation in the training dataset using Bartlett's test [7] for equal variance. Since the last PCs capturing random noise have equal variance they can be safely discarded. Because PCA is a global transformation of the dataset trying to minimize the global approximation error, local details are rarely represented by the first principal components and are therefore often neglected. Thus, PCA is good for a representation of overall trends and decomposition of the signal into different characteristics influenced by different driving forces, while ALA is good to represent localized information. How the decomposition of the signal into different characteristics can look like is shown in Figure 1(r). It is clearly visible, how the signal consists of an actual sensor response (visible in principal components one, two, and seven) and a superimposed beat, that is induced by sensor electronics and does not carry any relevant information for gas sensing.

3.3 Best Fourier Coefficients (BFC)

Depending on the nature of the measurement variable the relevant information might not be represented well in the

time domain. Especially in vibration analysis the information is expected to be much better represented in the frequency domain. To capture this information Fourier transformation is applied to transform raw data into the frequency domain. Extracted features are amplitudes and phase shifts. To reduce the dimensionality initially the symmetry property of the Fourier transformation is exploited which allows discarding half of the Fourier coefficients without loss of information. Then, only the ten percent of the coefficients with highest mean absolute value over all cycles are extracted as features. As described in [8], choosing these coefficients will preserve maximum signal energy for a reduction factor of ten. Note that using another reduction factor than ten enables the user to control the tradeoff between low approximation error and low number of extracted features. Figure 2(l) shows the resulting approximation when using BFC. As seen the overall cycle shape is more dominant than the superimposed beat (compare Figure 1(r)) and the beat is therefore discarded. The same accounts for the sharp edge at measurement point 80, that is represented by multiple high frequency coefficients with low amplitude.

3.4 Best Daubechies-Wavelet Coefficients (BDW)

Wavelet decomposition is applied to capture information in both the time and frequency domains. The Daubechies-4 Wavelet has been chosen because of its widespread use in signal processing and data compression. Wavelet decomposition is particularly interesting for providing

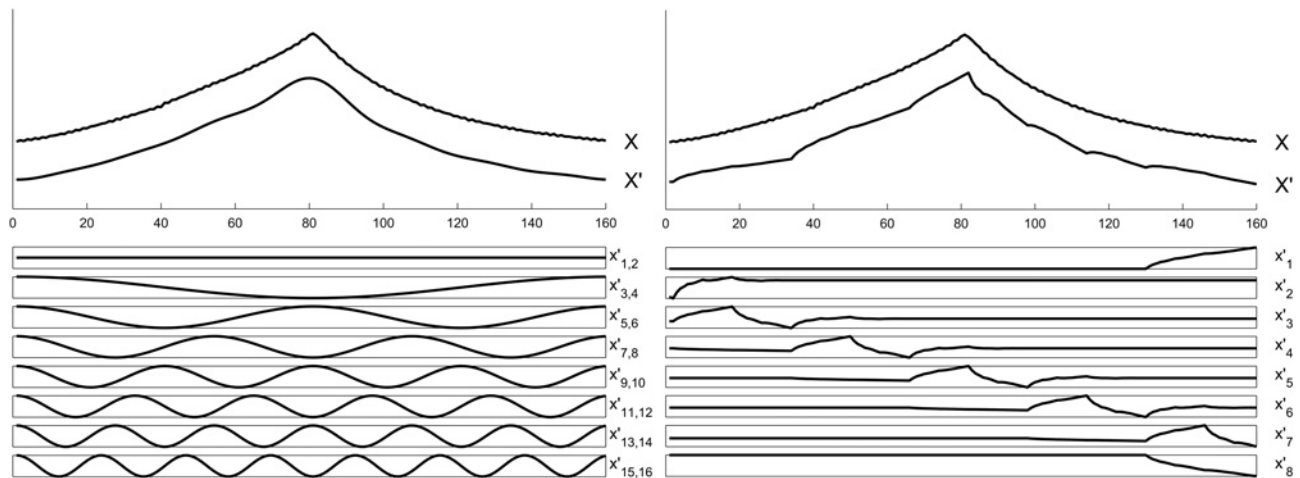


Figure 2: (l) Approximation X' using BFC with features $x'_1 - x'_{16}$ compared to the original gas sensor temperature cycle response X (shifted for better clarity). The approximation is the superposition of multiple sinus waves represented by amplitude (features with uneven indices) and phase shifts (features with even indices). (r) Approximation using BDW. Extracted features are wavelet coefficients $x'_1 - x'_8$.

a multiresolution view of the data since the first wavelet coefficients are extracted from global signal sections while the last coefficients are extracted from local details [8]. As for Fourier transformation the coefficients with highest mean absolute value over all cycles are extracted as features. Again, this method preserves the maximum signal energy for a given reduction factor and the reduction factor can be used to control the tradeoff between accuracy and number of features. The approximation using BDW is shown in Figure 2(l).

4 Feature selection

After feature extraction feature selection is applied to select the most relevant features for machine learning. As Houle et al. suggest [9] the “curse of dimensionality” is not caused by high dimensionality per se but by many features providing low contrast regarding classification. Thus, choosing only the most relevant features and discarding low contrast features efficiently suppresses the curse of dimensionality. Again, a huge variety of algorithms is available; however, none of these guarantees to find the optimal solution for all types of data. Therefore, the following, complementary set of algorithms is suggested:

4.1 Recursive Feature Elimination Support Vector Machines (RFESVM)

RFESVM is an embedded method for feature selection combining machine learning based on Support Vector Ma-

chines (SVM) with feature selection. The algorithm recursively trains a linear SVM on the current feature set and eliminates the feature with least relevance for group separation [10, 11]. As SVMs only address binary classification multiclass problems are resolved in the implementation using one vs. one multiclass encoding and averaging feature relevance over all binary classifiers. In addition, the parameter C of the binary SVMs, which controls the trade-off between training error and margin maximization (with $C = \infty$ leading to a hard margin SVM) [12], was set to a fixed value of 1,000 and all features are standardized before each SVM training. In addition, RFESVM was only applied to the 500 features with highest Pearson correlation to the target value to limit the computational effort.

4.2 Pearson correlation

Pearson correlation is used for quick results and feature pre-selection due to its low computational cost. Pearson correlation ranks features according to their individual linear correlation to the target. Both RFESVM and Pearson correlation select features for linear classification, but in contrast to Pearson correlation RFESVM takes into account feature interaction and any kind of monotonic relationship between features and target. This enables RFESVM to find more effective feature subsets and to eliminate redundancies that will occur when selecting the highest correlated features.

4.3 Univariate RELIEFF

For classes that are not linearly separable it seems natural to use learning algorithms based on nonlinear classification. RELIEFF is based on radial classification using k Nearest Neighbors (kNN). The implementation used is the univariate variant of the algorithm described in [13] and computes a relevance index based on kNN with $k = 5$. Since RELIEFF is based on radial classification it is complementary to RFESVM which is based on linear classification. However, the implementation of RELIEFF in this paper rates the relevance of features individually, unlike RFESVM which rates features in the context of other features. RELIEFF is applied to the 1,000 highest correlated features.

For all methods the optimal number of features is estimated by computing the tenfold cross-validation error of a Linear Discriminant Analysis (LDA) in combination with Mahalanobis classification as described in [14]. This means the group affiliation of a new point is predicted by assigning them to the group with minimum Mahalanobis distance to the point. Since LDA is computationally cheap the cross-validation error is computed for all feature sets with less than 500 features to select the number of features that yields minimum classification error. Note that any other classification algorithm that is suitable for the learning problem at hand could be used for the final classification step. The rather simple LDA is chosen here to demonstrate that dimensionality reduction does not increase model complexity and keeps the learning problem simple when applied as suggested in this paper.

5 Application of feature extraction and selection

The algorithms are combined by evaluating datasets with all combinations of the named feature extraction and selection algorithms as shown in Figure 3. This combination effectively finds the best features for linear and radial classification in time, frequency and time-frequency domain. Please note that this method, like every other one, does not guarantee finding the optimal feature representation. Nevertheless, the combination of complementary methods provides a high probability to find a relevant feature subset. To prove this concept and to show the applicability for different types of data the suggested set of methods was successfully applied to the following datasets from four different applications and an additional syn-

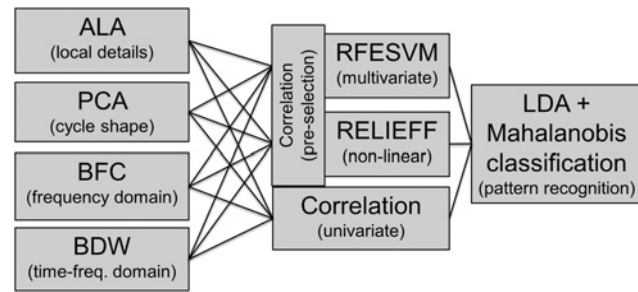


Figure 3: Schematic of the suggested algorithms for feature extraction (left), feature selection (middle) and classification (right) and their combination for automated dimensionality reduction and classification. Abbreviations: Adaptive Linear Approximation (ALA), Principal Component Analysis (PCA), Best Fourier Coefficients (BFC), Best Daubechies Wavelets (BDW), Recursive Feature Elimination Support Vector Machines (RFESVM), Linear Discriminant Analysis (LDA).

thetic data-set to verify the robustness of the algorithms against overfitting.

- Temperature cycled gas sensors [4, 14]
- Impedance spectroscopy of gas sensors [15]
- Condition monitoring using process sensors [1]
- Condition monitoring using vibration analysis [16]
- Random data

6 Results

Since this paper is focused on feature extraction and selection only results achieved using LDA are reported here. Since LDA is working best for classes forming single Gaussian distributed clusters low cross-validation errors reflect the success of representing data in an easily accessible way. All errors are reported as percentage of misclassified cycles based on tenfold cross-validation with LDA projection and consecutive classification using Mahalanobis distance to the class centers. Furthermore all the tested datasets have been evaluated using LDA in the respective paper, which makes it possible to trace all achieved results back to improved feature extraction and selection. Note that any other machine learning algorithm could have been used for classification.

6.1 Temperature cycled gas sensor

The first use case shown in this paper is the temperature cycled operation (TCO) of gas sensors. The goal here is to determine the concentration of hazardous naphthalene in the ppb range independent of varying ethanol background

as required for indoor air quality classification [14]. During training the gas sensor was exposed to six different concentrations of naphthalene, each in three different background concentrations of ethanol. A total of 1569 cycles with 160 data points, i.e. measurements of sensor conductance, per cycle were recorded [4]. The temperature cycled gas sensor can suffer from sensor drift and a varying concentration of background gas that strongly influences the sensor signal. These nonstationary effects prevent the application of time series modeling algorithms like Hidden Markov Models that require the signal to be stationary. Using the suggested approach allows the learning algorithm to take into account these effects and to compensate for them (with LDA only linearly).

For this dataset the choice of the feature extraction algorithm does not affect the classification performance significantly since all representations of the cycles lead to very low classification errors below 1%. However, feature selection is much more relevant to reduce the cross-validation error and reveal the basic structure of the data. This can be shown by comparing ALA applied with and without consecutive feature selection. When applied without feature selection the 20 features extracted by ALA lead to a cross-validation error of 5% while the best two features selected by RFESVM yield an error of 0.3%. This effect can be traced back to the fact, that the targeted naphthalene concentrations have been offered in three different background gases and do therefore not form Gaussian distributed clusters in feature space. Given the fact that LDA finds the low dimensional projection of the feature-space that maximizes the ratio of between class scattering

and within class scattering, LDA uses the additional features in the full set to minimize the within class scattering by projecting the within group clusters, that are formed by varying background, onto another. This effect increases overlap between different groups and is prevented by feature selection since feature selection removes the feature that allow LDA to misfit the sample distribution. The corresponding two-dimensional LDA-projections can be found in Figure 4. Although this problem is caused by the classifier used and could be circumvented using a nonlinear classifier it shows how feature selection can simplify a learning problem which comes in handy when applied to a high dimensional problem that cannot be solved directly with a nonlinear classifier.

6.2 Impedance spectroscopy of gas sensor

In the second use case impedance spectroscopy was applied to a gas sensor to discriminate different gases and to simultaneously assess the sensor's degradation state that is increased by several stages throughout the measurement by sensor poisoning. The goal is to find a classifier that is capable of determining gas type and degradation state. A total of 7983 complex spectra containing measurements of admittance at 801 different frequencies were recorded [15]. For dimensionality reduction each spectrum is treated as one cycle consisting of two sensors measuring real and imaginary part of the admittance. Feature extraction is applied to each sensor separately. Afterwards features extracted by the same method are grouped

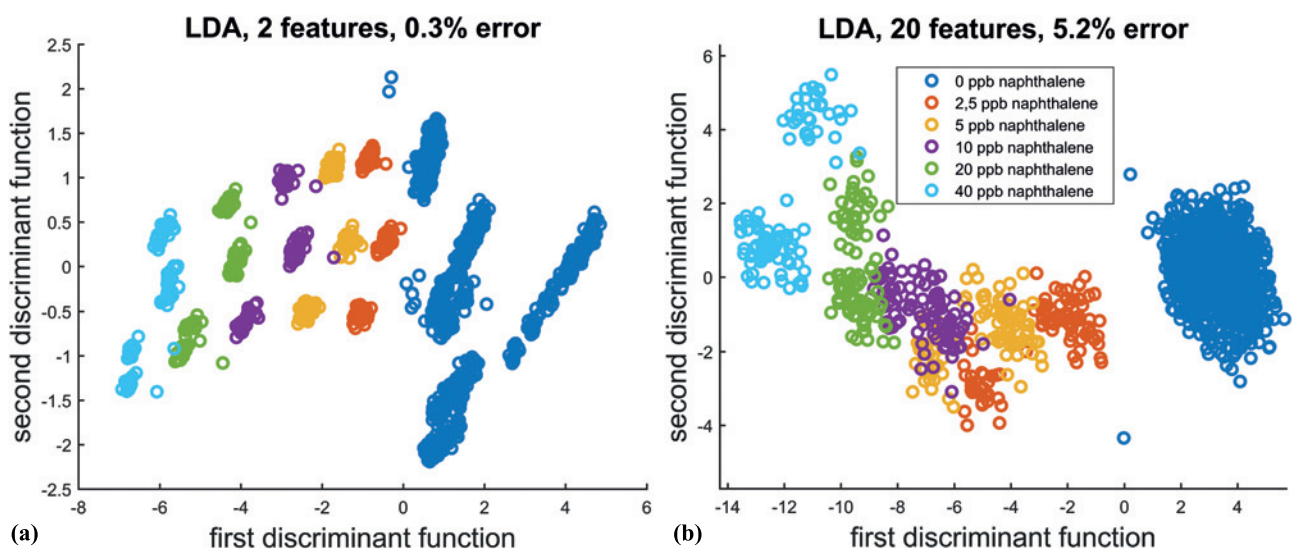


Figure 4: (a) LDA scatter plot of TCO gas sensor data using two most relevant features. (b) LDA scatter plot of TCO gas sensor using 20 features. Abbreviations: Linear Discriminant Analysis (LDA), parts per billion (ppb).

together before feature selection is applied. This approach is used for all discussed datasets that contain more than one sensor. For more details see [15].

For this problem the best classification results were achieved using a combination of ALA and RFESVM. The reported classification error is 1.8%. This result shows that the proposed set of methods can easily be applied to evaluate data coming from multiple sensors. Furthermore this shows the applicability to spectral data. It is worth noting, that the classification error increases to 4.6% when feature extraction is performed on sensor impedance instead of admittance. The fact, that impedance and admittance contain the exact same information but yield different classification errors, shows the importance of data representation for machine learning. This emphasizes the importance of feature extraction and selection, which can be further increased by improving the raw data by preprocessing.

6.3 Condition monitoring using process sensors

In the third use case machine learning is applied to classify the progression of different fault mechanisms of a complex hydraulic system. During training the hydraulic machine is operated with a fixed work cycle. The setup of the test system enables the operator to simulate different fault mechanisms like accumulator pressure decrease in different states of progression. During training 1,449 cycles with multiple simulated faults were recorded. A total of 15 process sensors for pressure, temperature, flow, electrical power and vibration were used to generate signals between 60 and 6,000 measurement points per cycle. The goal is to classify the progression of a single fault-mechanism independent of all other fault-mechanisms. For more details see [1].

In the previous work by Helwig et. al. [1] the working cycle of the hydraulic machine was split into thirteen segments that each cover a static or transient part of the cycle. Subsequently from all segments of all sensor signals the features median, variance, slope, position of maximum, skewness and kurtosis are extracted. From this feature set the 20 features with highest absolute Pearson correlation to the respective target are selected for LDA training. In comparison with this method the automated approach achieved improvements of the classification rate while simultaneously the manual effort for feature extraction was significantly reduced. Regarding classification performance the highest increase was achieved for accumulator pressure of the hydraulic system. The previously reported error rate of 9.6% [1] was decreased to only 0.35%

using a combination of ALA and RFESVM. There was no significant improvement in classification performance for the other possible fault mechanisms since the previously achieved results were already very good preventing significant further improvement. Nevertheless, in these cases the effort required for feature extraction by manually defining the cycle segments from which features are extracted was greatly reduced by ALA automatically identifying linear segments. The good results achieved on this dataset show that the statistical approach is still applicable if the machine is too complex for physical modeling.

6.4 Condition monitoring using vibration analysis

The last use case is vibration analysis for condition monitoring of the hydraulic system described above. Instead of process sensors an accelerometer at the main oil pump was used to record vibrations during the 975 working cycles with 483,328 points per cycle. For more details see [16].

The application of the described set of features to vibrational data both shows the scalability of the methods to nearly half a million measurement points per cycle and the applicability for information best represented in frequency and time-frequency domains. Furthermore it shows that the best data representation depends on the respective training target, i.e. the fault mechanism to be quantified. For example the decrease in cooling power is better represented by raw time domain data (perfect classification using highest correlated Daubechies-4 Wavelet coefficients) than by the amplitude spectrum after Fourier transformation applied as preprocessing (25% error using highest correlated Daubechies-4 Wavelet coefficients). On the other hand, the decrease in accumulator pressure is better represented in the frequency domain (0% error using highest correlated Daubechies-4 Wavelet coefficients of the amplitude spectrum vs. 17.2% error using highest correlated Fourier coefficients of time domain data).

6.5 Random data

The feature selection step is not included in cross-validation although being supervised and therefore being potentially prone to overfitting, i.e. possible overfitting in feature selection will not be suppressed by cross-validation of the classifier. To estimate the amount of overfitting the set of algorithms explained above is applied to randomized data. This dataset was created by randomly shuffling

cycles and measurement points of the TCO gas sensor measurement and assigning each cycle randomly to one of three classes. The cross-validation errors of all combinations of the four feature extraction and two feature selection algorithms described above are distributed around the mean value of 66% with a standard deviation of 1.5%. The expected value for randomly guessing class membership of three classes based on random data is 66.7%; thus, the amount of overfitting in feature selection is small but might increase for larger datasets. Nevertheless feature selection is not cross-validated to save computation time and to make results comparable to results reported in previous publications. Cross-validation of feature selection is a work in progress.

7 Conclusion

As demonstrated the suggested set of methods can be applied to a wide range of different applications. In all four use cases considered in this paper at least one of the suggested methods was able to capture the essential information contained in the cyclical data leading to excellent and often even perfect classification results despite the rather simple classification algorithm used. This again emphasizes the importance of meaningful data representation. Like every other method, this set of methods does not guarantee finding a good data representation but offers a high probability to work well for real world datasets. Nevertheless it is a useful tool for predictive maintenance of complex machines and plants based on sensor networks or development of smart sensor systems compensation disturbing interferences like aging or poisoning of sensor element.

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